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Abstract

Machine learning and deep learning have demonstrated remarkable success across a wide range of time series tasks. However, their lack of transparency limits explainability and trust. To address this challenge, numerous explanation methods have been proposed in recent years, each of them providing distinct perspectives on how to interpret time series models through signals, patterns, structures, or semantics. This survey synthesizes these perspectives into a unified cognitive hierarchy that bridges methodological diversity with human understanding. Existing approaches are classified into four levels: signal-space, pattern-space, structural, and semantic, reflecting a progressive shift from low-level signal attribution to high-level semantic reasoning. Building on this taxonomy, we conduct comprehensive literature review, revealing cross-level connections, emerging trends, and critical gaps in current research. Our analysis underscores the growing importance of semantic explanations, particularly those facilitated by large language models (LLMs), as a promising direction for achieving human-centered and readable explainability. In addition, we compile and categorize data benchmarks used for explaining diverse time series tasks and provide an accompanying open repository to support continued research in this field.

From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens

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Machine learning and deep learning have demonstrated remarkable success across a wide range of time series tasks. However, their lack of transparency limits explainability and trust. To address this challenge, numerous explanation methods have been proposed in recent years, each of them providing distinct perspectives on how to interpret time series models through signals, patterns, structures, or semantics. This survey synthesizes these perspectives into a unified cognitive hierarchy that bridges methodological diversity with human understanding. Existing approaches are classified into four levels: signal-space, pattern-space, structural, and semantic, reflecting a progressive shift from low-level signal attribution to high-level semantic reasoning. Building on this taxonomy, we conduct comprehensive literature review, revealing cross-level connections, emerging trends, and critical gaps in current research. Our analysis underscores the growing importance of semantic explanations, particularly those facilitated by large language models (LLMs), as a promising direction for achieving human-centered and readable explainability. In addition, we compile and categorize data benchmarks used for explaining diverse time series tasks and provide an accompanying open repository¹ to support continued research in this field.

Key Words: Time Series Tasks, Explainable AI, Time Series Explainability

1 Introduction

Time series are widely applied in many real-world scenarios, including finance [105, 155], healthcare [66, 127, 142], environmental science [2, 162], and transportation [93, 141], as illustrated in Figure 1. With the rapid advance of deep learning technologies, time series models have achieved remarkable predictive performance. However, the increasing complexity and opacity of these models have raised significant concerns about explainability and transparency, especially in critical and high-stake domains. In domains where decisions impact financial stability or human lives, unexplainable model behaviors may lead to serious risks, such as financial losses or misdiagnoses [121, 134]. A lack of explainability can undermine users' trust and hinder practical adoption of deep models, as their predictions often lack explainable reasoning that users can understand or validate. In contrast to static data, such as images and text [129, 132], time

¹<https://github.com/realMoana/awesome-time-series-explanation-papers>

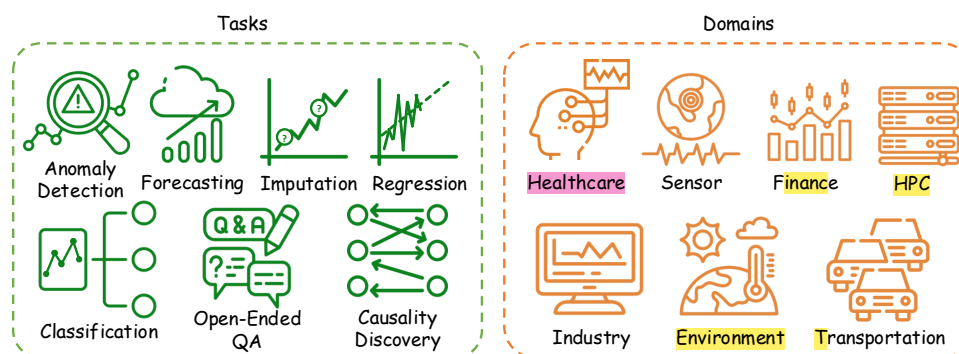


Fig. 1. Applications of time series across multiple tasks and domains.

series exhibit complex temporal dependencies, non-stationarity, and cross-feature correlations, making their internal reasoning process even more challenging to explain.

To tackle these issues, a wide range of time series explanation methods have been proposed in recent years [61, 76, 89, 123, 166], aiming to uncover the decision rationale of deep models and enhance human understanding of temporal dynamics. While these methods brought significant advances, most of them focus on specific explanation aspects, such as assigning importance to individual time steps, visualizing salient subsequences, identifying representative temporal prototypes, or generating textual explanations [61, 75, 86, 112, 140]. These approaches provide valuable insights at certain levels of explanation. However, they remain model-centered and limited in scope, focusing on how to explain model behavior internally rather than how such explanations correspond to human perception and reasoning. This disconnection leads to a persistent misalignment between model explanations and human interpretive understanding. Existing taxonomies [104, 147, 180] may have consolidated this gap by categorizing methods via algorithmic design or model dependency, failing to capture the progressive nature of human cognition across different levels of abstraction.

To bridge this gap, we introduce a novel human-cognitive hierarchy perspective that structures time-series explainability into four distinct levels: *signal-space*, *pattern-space*, *structural*, and *semantic*, as illustrated by Fig. 3. This framework aligns computational explainability with human cognitive processes, illustrating a progression from low-level signal reasoning to high-level semantic understanding. Guided by this framework, our survey systematically reviews existing methods and evaluation strategies across these four levels, providing a unified synthesis of how time series explanations are generated. To summarize, the key contributions of this work include:

- We propose a human-centric taxonomy that structures time series explainability by cognitive depth through a four-level hierarchy, providing a coherent framework for understanding and comparing existing studies in terms of their *explanation granularity*, *methodological design*, and *human explainability*.
- Building on this taxonomy, which defines four cognitive levels, namely signal-space, pattern-space, structural, and semantic, we systematically review the literature to reveal hidden trends, cross-level connections, and research gaps.
- We highlight the emerging role of semantic explanations, especially those enabled by Large Language Models (LLMs), as a crucial step toward making complex time series models understandable to non-expert users.
- Our analysis reveals developmental imbalances across these levels: while pattern-level explanations have reached a mature stage, semantic-level reasoning remains underexplored. This perspective provides a clear roadmap and actionable insights for guiding future research in time series explainability.

Table 1. Comparison of Our Survey with Related Time Series Explainability Surveys

		Rojat et al.[130]	Guidotti et al.[45]	Theissler et al.[147]	Mochaourab et al.[104]	Zhao et al.[180]	Arsenault et al.[5]	Our Survey
Task	Classification	✓	✓	✓	✓	✓		✓
	Forecasting	✓				✓	✓	✓
	Regression					✓		✓
	Imputation							✓
	Anomaly Detection					✓		✓
	Causality Discovery							✓
	Question Answering							✓
Method	Signal-space	✓		✓	✓	✓	✓	✓
	Pattern-space	✓	✓	✓	✓	✓		✓
	Structural						✓	✓
	Semantic							✓
Evaluation	Qualitative	✓		✓		✓		✓
	Quantitative	✓	✓	✓		✓	✓	✓
Dataset Summary								✓
Count of Explanation Types		7	1	7	3	8	4	15

- We compile and categorize a comprehensive collection of datasets used for explaining different types of time series tasks, providing a unified reference for benchmarking.

Related Surveys and Novelty of This Survey. Previous surveys on time series explainability have taken limited perspectives. Theissler et al. [147] focus on explanation models for time series classification tasks. Guidotti et al. [45] review methods for generating counterfactual explanations, though not specific to time series data. Mochaourab et al. [104] only summarize post-hoc explanation methods. Rojat et al. [130], Zhao et al. [180], and Arsenault et al. [5] propose taxonomies based on the ante-hoc versus post-hoc distinction. In particular, Arsenault et al. [5] focus on forecasting tasks in the financial domain. However, these surveys do not capture how explanation forms evolve toward human-understandable reasoning. This survey organizes time series explainability through the lens of human cognitive hierarchy. This perspective moves beyond traditional method-driven categorizations and provides a unified framework that connects low-level signal attributions to high-level semantic reasoning. This survey covers classification, forecasting, anomaly detection, regression, and other tasks, which is more comprehensive than existing surveys. Table 1 presents a comparison against related surveys, including task coverage, method level, evaluation, dataset summary, and the number of explanation types covered.

The rest of the paper is organized as follows. Section 2 introduces fundamental concepts and definitions related to time series models. Section 3 focuses on time series explanation methods and categorizes them according to our proposed taxonomy, and also summarizes evaluation metrics for assessing explanation quality. Finally, Section 5 discusses the key challenges and promising future research directions. Figure 2 provides an overview of this survey’s structure.

2 Background and Problem Formulations

2.1 Notations

A time series $X \in \mathbb{R}^{l \times d}$ is an ordered set of l time steps with d features. When $d = 1$, each time step is associated with a single scalar observation, $X = [x_1, x_2, \dots, x_l]$, $x_i \in \mathbb{R}$, which corresponds to a **univariate time series** (UTS). When $d > 1$, each time step contains d -dimensional observation vector, $X = [x_1^\top, x_2^\top, \dots, x_l^\top]^\top$, $x_i \in \mathbb{R}^d$, which corresponds to a **multivariate time series** (MTS).

When time series are collected with heterogeneous multimodal data, such as text, image, and structural information, they form a **multi-modal time series**. Additional modalities provide complementary context: 1) *Text*. It includes static

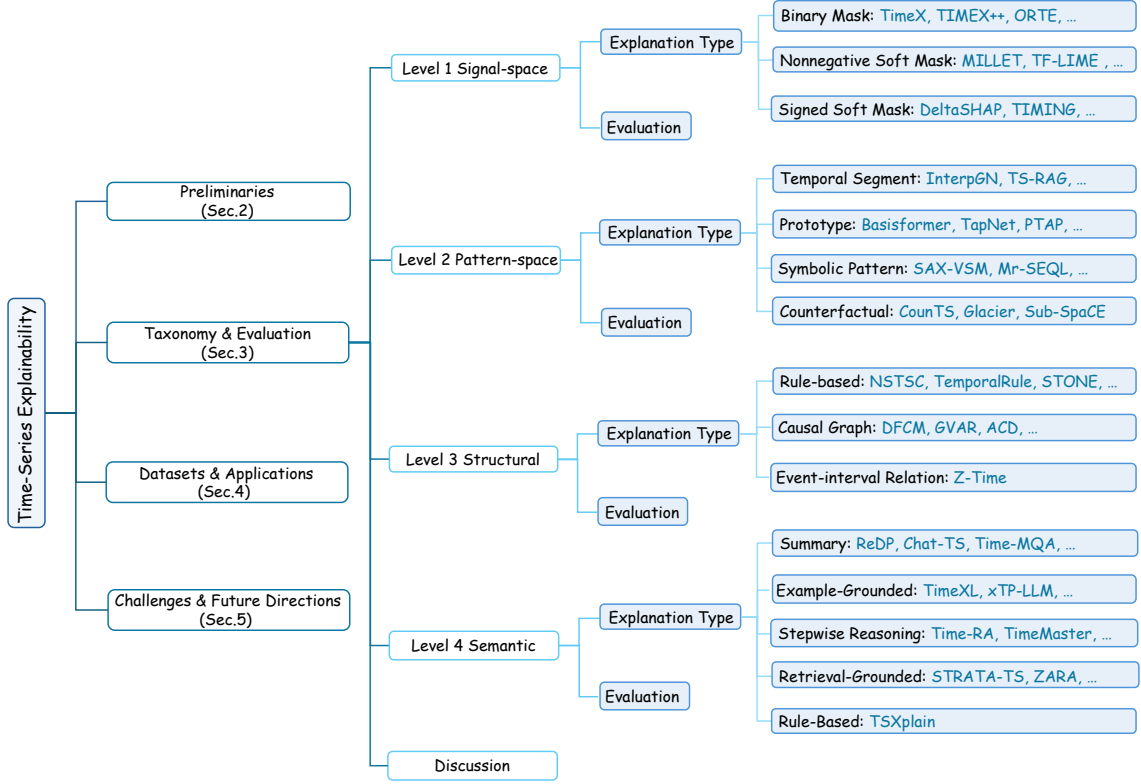


Fig. 2. Structure overview. This survey is structured around a four-level cognitive framework for time series explainability, including signal-space, pattern-space, structural, and semantic explanations. It is supported by sections on preliminaries, taxonomy and evaluation, datasets and applications, and future directions.

textual information — such as domain-level summaries [51] or patient profiles [126] — and dynamic textual information — such as time-stamped clinician notes [75] or system log messages [48]. 2) *Image*. Static image information consists of images associated with the observed entity, including medical images [176] or snapshots [64]. Dynamic image information comprises time-ordered image sequences, such as CT slices acquired during disease progression [173] or satellite imagery collected at regular intervals [79]. 3) *Structure*. This modality captures relationships among entities or variables. The structure may be static — such as in fixed road networks [61] or Internet of Things (IoT) device layouts [83] — or dynamic, reflecting evolving systems — such as changing traffic connectivity [25] or evolving correlations in financial markets [17]. Details of multi-modal time series will be discussed in Section 4.1.

2.2 Time Series Tasks

Time Series Classification. Given a dataset $\mathcal{D} = \{(X^{(i)}, y^{(i)})\}_{i=1}^N$ containing N pairs, where $X^{(i)}$ denotes the time series instance and $y^{(i)}$ the corresponding labels. Each label $y^{(i)}$ belongs to one of the classes in $C = \{1, 2, \dots, c\}$. Time series classification is the task of training a classification model $f: \mathbb{R}^{l \times d} \mapsto C$ which maps an input time series $X^{(i)}$ to its predicted label.

Time Series Forecasting. Given a historical time series $X \in \mathbb{R}^{l \times d}$, where l is the look-back window size. The goal of forecasting is to predict future values of the sequence $\hat{Y} \in \mathbb{R}^{h \times d}$, which start at time step $l + 1$, based on the past observations in X . Formally, we aim to learn a predictor $f : \mathbb{R}^{l \times d} \mapsto \mathbb{R}^{h \times d}$ that maps the historical sequence X into its future values $\hat{Y}$ over a horizon h .

Time Series Regression. Given a dataset $\mathcal{D} = \{(X^{(i)}, y^{(i)})\}_{i=1}^N$, where $X^{(i)}$ is a time series and $y^{(i)}$ is a continuous target, the goal is to learn a regression model $f : \mathbb{R}^{l \times d} \mapsto \mathbb{R}$ that maps temporal observations to real-valued predictions. Regression differs from classification in that it predicts a continuous value for a variable external to the input time series instead of a categorical value. It also differs from forecasting, which predicts the future values of the time series itself, whereas regression predicts an externally defined numerical outcome for the entire input sequence.

Time Series Imputation. For a time series $X \in \mathbb{R}^{l \times d}$ with missing values, the goal is to recover the missing values using the observed portions. Define a binary mask $M \in \mathbb{R}^{l \times d}$, where $M_{t,j} = 0$ if the j -th feature of step t , $X_{t,j}$, is missing, and $M_{t,j} = 1$ otherwise. Let $\hat{X} \in \mathbb{R}^{l \times d}$ denote the imputed values. The imputed time series is $X_{\text{imputed}} = X \odot M + \hat{X} \odot (1 - M)$.

Time Series Anomaly Detection. Given a time series X , anomaly detection aims to identify abnormal time steps that deviate from normal behavior. The model f assigns anomaly scores $\mathbf{a} = (a_1, \dots, a_l)^\top \in \mathbb{R}^l$ to determine which time step in X is anomalous. The final decision is made by comparing each score a_i with a threshold δ : anomalous if $a_i > \delta$ and normal otherwise.

Time Series Causal Discovery. Given an MTS X , the goal is to infer a directed causal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ that encodes which variables influence others in a causal sense, where variables $\mathcal{V}$ and directed causal edges $\mathcal{E}$ are inferred from the information in X , such as temporal dependencies or interventions.

Time Series Open-Ended Reasoning & QA. With the rise of foundation models, time series can be paired with natural language queries q for reasoning or question answering. The goal is to learn a function $f(X, q)$ that produces an answer, which may be textual, numerical, or symbolic.

2.3 Time Series Explainability

Given a dataset $\mathcal{D} = \{(X^{(i)}, y^{(i)})\}_{i=1}^N$, the goal of time series explanation is to construct a human-understandable representation that reveals how a predictive model f transforms the input $X^{(i)}$ into the output $y^{(i)}$. Formally, an explanation can be viewed as a mapping $\phi(X^{(i)}) = Z^{(i)}$, where $Z^{(i)}$ denotes an explanation, possibly in different forms, such as salient signals, representative patterns, causal structures, or textual rationales. Based on the dependency between the time series prediction task and the explainability mechanism, explanation methods are divided into two classes: *ante-hoc* and *post-hoc*.

Ante-hoc Explanations. In ante-hoc methods, time series prediction function can be structured as $f = g \circ \phi$, where the explanation function ϕ is an integral component of the model architecture, and $f = g(X^{(i)}, \phi(X^{(i)})) = \hat{y}^{(i)}$ produces the final prediction. Both g and ϕ are jointly optimized during training. Attention-based methods are typical examples of this paradigm [32, 39, 53, 118], where the attention weights serve as interpretable importance indicators.

Post-hoc Explanations. In post-hoc methods, the explanation function ϕ is derived after training a prediction model f . The explanations depend critically on information extracted from the pretrained function f , such as gradients [145] or intermediate activations [132]. Gradient-based attribution methods [37, 151] are examples in this category.

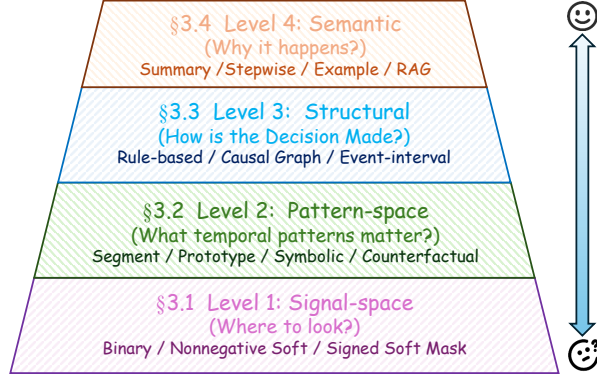


Fig. 3. Taxonomy of explanation types. We structure time series explanations into four cognitive levels. *Signal-level* informs where to look from the input signals; *pattern level* identifies what patterns matter to the predictions; *structural level* tells how a decision is made; *semantic level* explains why a predicted result happens. A higher level provides explanations that more closely reflect human cognition than a lower level.

3 Taxonomy

In this section, we review explanation methods designed for time series tasks and organize them according to our proposed taxonomy as illustrated in Figure 3. Time series explanation methods vary significantly in their assumptions, learning objectives, and representation forms [22, 86, 112, 140]. Despite this variety, they share a common goal: uncovering the reasoning process of temporal models on sequential data and making this reasoning transparent and intelligible to humans. Our taxonomy provides a structured view of this domain by categorizing existing methods into four levels: **signal-space**, **pattern-space**, **structural**, and **semantic**. Each level represents a distinct depth of understanding. At the signal-space level [89, 160], explanations focus on local feature contributions over time. In the pattern-space level [63, 122], they highlight recurring temporal patterns. At the structural level [46, 69], explanations capture cross-variable and hierarchical dependencies, and at the semantic level [70, 73], they connect numerical representations with high-level human concepts. This organization reflects how people perceive and reason about time-related events, evolving from raw signals to informative semantics. It provides a foundation for systematically analyzing existing techniques. Following this taxonomy, Sections 3.1–3.4 describe the main approaches at each level and discuss representative models and evaluation methods.

3.1 Where to Look: Signal-space Level

The signal-space level represents the most fundamental layer of abstraction in time series explanation. At this level, explanatory relevance connects with the original input signals in either the temporal or frequency domain. The model’s decision is explained by local contribution of individual signal components, which directly reflects its influence in the final prediction.

The goal [110] is to identify the most informative part of the input $\tilde{X} = X \odot M$ that satisfies:

$$\|I(X, y) - I(\tilde{X}, y)\| \leq \epsilon, \quad \text{s.t.} \quad \|M\| \leq \eta, \quad (1)$$

where $I(\cdot, \cdot)$ denotes mutual information, ϵ, η are small constants, $\odot$ denotes element-wise multiplication, and $M \in \mathbb{R}^{l \times d}$ is a timestep-wise importance indicator, $M_{t,j}$ indicates the importance weight assigned to the j -th feature at time step t .

Table 2. Key highlights of signal-space explanations. Abbreviations: C = Classification, F = Forecasting, R = Regression, OIR = Optimal Information Retention, IB = Information Bottleneck, MI = Mutual Information, LRP = Layer-wise Relevance Propagation.

Model	Mask Type	Methodology	Task	Ante-hoc/Post-hoc
ORTE [175]	Binary	OIR	Classification	Post-hoc
TIMEX++ [89]	Binary	IB	Classification	Post-hoc
IndMask [110]	Binary	MI	C/F	Post-hoc
TimeX [123]	Binary	Surrogate	Classification	Post-hoc
ContraLSP [90]	Binary	Perturbation	C/R	Post-hoc
Meng2023 [102]	Binary	Perturbation	Classification	Post-hoc
TF-LIME [160]	Nonnegative Soft	Perturbation	Classification	Post-hoc
SpectralX [23]	Nonnegative Soft	Perturbation	Classification	Post-hoc
WinIT [77]	Nonnegative Soft	Perturbation	Classification	Post-hoc
Extrmask [33]	Nonnegative Soft	Perturbation	Classification	Post-hoc
TSInsight [137]	Nonnegative Soft	Perturbation	Classification	Post-hoc
Dynamask [24]	Nonnegative Soft	Perturbation	C/R	Post-hoc
Series Saliency [117]	Nonnegative Soft	Perturbation	Forecasting	Post-hoc
StartGrad [151]	Nonnegative Soft	Gradient	Classification	Post-hoc
XCM [37]	Nonnegative Soft	Gradient	Classification	Post-hoc
MTEX-CNN [8]	Nonnegative Soft	Gradient	C/F	Post-hoc
MILLET [32]	Nonnegative Soft	Attention	Classification	Ante-hoc
ETN-ODE [39]	Nonnegative Soft	Attention	Forecasting	Ante-hoc
LAXCAT [53]	Nonnegative Soft	Attention	Classification	Ante-hoc
TFT [85]	Nonnegative Soft	Attention	Forecasting	Ante-hoc
TCN+Attention [118]	Nonnegative Soft	Attention	Forecasting	Ante-hoc
DSAN [86]	Nonnegative Soft	Attention	Forecasting	Ante-hoc
IMV-LSTM [49]	Nonnegative Soft	Attention	Forecasting	Ante-hoc
RETAIN [22]	Nonnegative Soft	Attention	Classification	Ante-hoc
DeLELSTM [156]	Nonnegative Soft	Decomposition	Forecasting	Ante-hoc
mWDN [159]	Nonnegative Soft	Decomposition	C/F	Ante-hoc
FLEXtime [12]	Nonnegative Soft	Filterbank	Classification	Post-hoc
TIMING [57]	Signed Soft	Integrated Gradients	Classification	Post-hoc
DeltaSHAP [68]	Signed Soft	Feature Permutation	Classification	Post-hoc
ShapTST [18]	Signed Soft	Shapley Estimation	C/R	Ante-hoc
DFT-LRP [154]	Signed Soft	LRP	Classification	Post-hoc
ShapTime [179]	Signed Soft	Cooperative Game	Forecasting	Post-hoc
DEMUX [30]	Signed Soft	Perturbation	Classification	Post-hoc
TimeSHAP [11]	Signed Soft	Perturbation	Classification	Post-hoc
PERT [119]	Signed Soft	Perturbation	Classification	Post-hoc
FIT [149]	Signed Soft	Distribution	Classification	Post-hoc

The objective of such explanation methods is to learn a function $\phi(X) = M$ that can capture fine-grained importance distributions over the original signal and reveal where the model attends most during inferences. Table 2 lists representative signal-space explanation methods.

3.1.1 Explanation Types. These methods fall into three categories according to how mask values are defined: *binary masks*, *nonnegative soft masks*, and *signed soft masks* [23, 123, 179]. For a time series X and a mask M , which is applied element-wise as $M \odot X$, a binary mask is denoted as $M \in \{0, 1\}^{l \times d}$, a nonnegative mask is denoted as $M \in [0, 1]^{l \times d}$, and a signed mask is denoted as $M \in [-1, 1]^{l \times d}$. These masks are visualized in Figure 4.

Binary Mask. Binary mask methods identify critical time steps or features through discrete selections, highlighting the input regions that contribute most to model predictions. As illustrated in Fig. 4(b)(i), by producing sparse and unambiguous attributions, they enable a clear separation between relevant and irrelevant temporal components, which is particularly valuable in high-stakes applications such as clinical monitoring [136] and fault detection [167], where explanations must be concise and verifiable.

Perturbation-based methods [90, 102] derive sparse binary masks by perturbing the input and observing the resulting prediction shifts. Through generative optimization, Meng et al. [102] identify the smallest subset of features that support

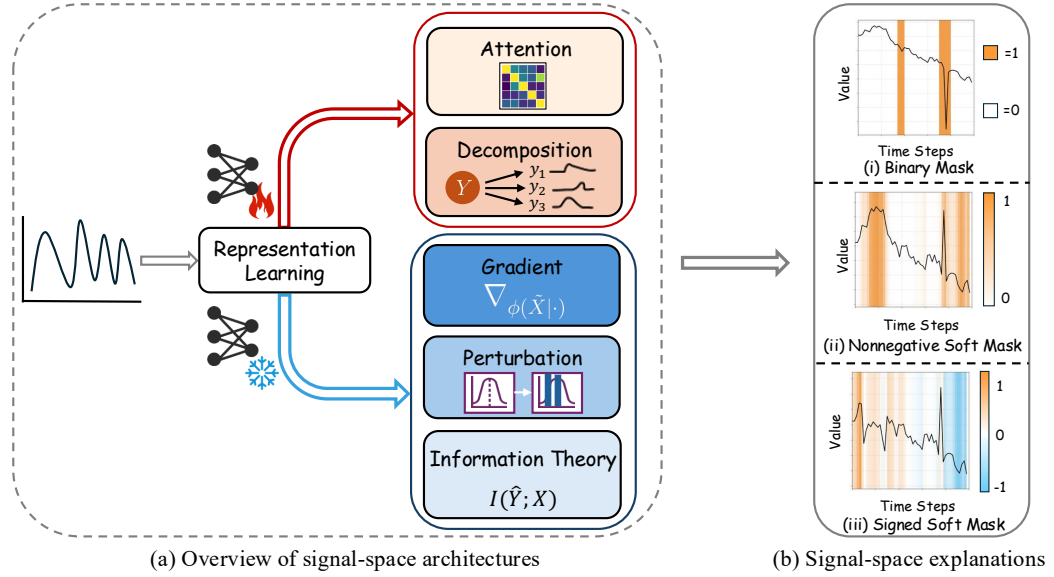


Fig. 4. Illustration of signal-space methods. (a) is an overview of signal-space architectures. These methods get explanations through attention mechanisms, decomposition modules, gradients, perturbation strategies, or information-theoretic criteria. (b) shows the types of signal-space explanations.

a given decision, while ContraLSP [90] integrates contrastive learning with sparse gating mechanisms. For information-theoretical frameworks, IndMask [110] maximizes the mutual information (MI) between model predictions and masked inputs, using a cross-entropy loss to approximate the conditional entropy term. TIMEX++ [89] and ORTE [175] extend the same information-theoretical principle under the Information Bottleneck (IB) [148] and Optimal Information Retention (OIR) [175] principles, respectively, aiming to balance predictive information and redundancy from the original input. In addition, by training a surrogate model, TimeX [123] generates binary masks that preserve predictions while enforcing sparsity.

Nonnegative Soft Mask. In contrast to binary masks, soft mask methods (Fig. 4(b)(ii)) relax the discrete constraint and learn differentiable masks whose values represent graded importance. Nonnegative soft mask methods assign each timestep or feature an attribution score that reflects its contribution magnitude to the model prediction, higher scores indicate more influential inputs.

Perturbation-based masks estimate regional influence by masking, removing, or replacing input segments and observing the induced prediction changes. Series Saliency [117] applies the smallest destroying region (SDR) principle [26] to find minimal subsequences whose removal most reduces the prediction score. With an autoencoder-based perturbation module, TSInsight [137] reconstructs important regions while diminishing the impact of less significant ones, which improves regional explainability. Dynamask [24] optimizes smooth real-valued masks with sparsity regularization. Through continuous perturbation learning, Extrmask [33] refines mask optimization and improves temporal smoothness. WinIT [77], SpectralX [23], and TF-LIME [160] perturb time or time-frequency patches and measure importance based on the resulting prediction changes. Gradient-based methods calculate saliency from input gradients or gradient-weighted activations. MTEX-CNN [8] generates input gradient maps that measure local sensitivity. By

weighting intermediate activations with class gradients, XCM [37] produces Grad-CAM style [132] relevance maps that reveal temporal importance. StartGrad [151] stabilizes gradient attributions through normalization and smoothing to improve calibration across timesteps and features. Attention-based models capture temporal and feature relevance through learned attention weights. RETAIN [22] applies visit-level and variable-level attention in reverse time to produce instance-level explanations. IMV-LSTM [49] separates intra-variable and inter-variable attention to distinguish feature and temporal relevance. TFT [85] combines explainable multi-head attention with variable-selection gates for multivariate forecasting. TCN+Attention [118], DSAN [86], LAXCAT [53], ETN-ODE [39], and MILLET [32]) use a similar mechanism to generate nonnegative attention weights. Finally, through a learnable filterbank [152], FLEXtime [12] evaluates spectral importance over time, integrating frequency-aware explainability.

Signed Soft Mask. Signed soft mask methods assign attribution scores that can take both positive and negative values, which can be observed in Fig. 4(b)(iii). This indicates whether each timestep or feature supports or undermines the model’s prediction for the target class.

For example, FIT [149] evaluates feature importance through shifts in distributions when selected inputs are replaced or removed, capturing their statistical effect on the model output. Additionally, perturbation-based methods [11, 30, 119] estimate feature relevance by masking, removing, or replacing input components and monitoring the resulting prediction changes: PERT [119] learns replacement-based perturbation masks through gradient-guided optimization to find minimal but influential feature subsets; following a Shapley-based formulation, TimeSHAP [11] sequentially removes features to estimate marginal temporal contributions, providing order-consistent attributions; and DEMUX [30] extends TimeSHAP’s perturbation framework with gradient guidance to produce class-specific explanations. By modeling time steps as cooperative players, ShapTime [179] calculates fair attributions based on coalition interactions, offering an explainable view of temporal cooperation. On the other hand, DeltaSHAP [68] estimates feature contributions by measuring prediction differences before and after feature perturbations, efficiently approximating changes in Shapley values. ShapTST [18] also incorporates Shapley-based pre-training into time series transformers, allowing single-pass estimation of Shapley-value-based contributions from features and time steps. Layer-wise relevance propagation (LRP)-based explainers, such as DFT-LRP [154], extend relevance propagation to the frequency domain, tracing prediction relevance through spectral representations. TIMING [57] derives explanations by path-integrating gradients from a baseline to the input, using a Riemann-sum approximation to obtain importance scores.

3.1.2 Evaluation. The evaluation of signal-space explanations aims to evaluate how accurately and consistently the methods reflect a model’s true decision process. To achieve this, both qualitative assessments and quantitative metrics are incorporated to examine explanation fidelity and robustness. **Qualitative evaluation** focuses on quality through controlled user studies, whereas **quantitative metrics** provide objective measures of how well the highlighted regions align with ground-truth annotations, preserve predictive performance, or remain stable under perturbations, with metrics covering **ground-truth-based evaluation**, **faithfulness**, **complexity**, and **robustness**. These approaches offer a comprehensive understanding of the explanatory reliability and practical value of signal-space methods.

Qualitative Evaluation. At the signal-space level, the explainability of temporal heatmap explanations is occasionally evaluated through controlled user studies. One such instance is reported in [23], where participants with machine learning backgrounds were asked to compare class-wise explanation maps generated by several methods. Each participant was shown class-averaged signals with top-ranked important regions highlighted and asked to judge which explanation most clearly captured the distinctive temporal pattern of each class. The aggregated rankings were used to quantify human-perceived explanation quality.

Ground-truth Based Evaluation. When ground-truth annotations are available, precision–recall metrics serve as direct indicators of how accurately an explanation identifies truly important time steps or features. Common measures include the Area Under Precision (AUP), Area Under Recall (AUR), and Area Under the Precision–Recall Curve (AUPRC) [12, 89, 123, 160, 175]. Alternative formulations such as the Area Over the Perturbation Curve to Random (AOPCR) and the Normalized Discounted Cumulative Gain at n (NDCG@ n) [32] capture the ranking consistency between predicted saliency and true feature relevance. Intersection over Union (IoU) [30, 37, 160] measures the overlap between highlighted and annotated regions, whereas Rank-Biased Overlap (RBO) [23] measures the alignment between the ranked lists of predicted and true salient regions, assigning greater weight to higher-ranked features.

Faithfulness. Since real-world datasets typically lack ground-truth annotations, many studies evaluate explanation quality indirectly by observing model behavior after perturbations. Accuracy Drop and Cross-Entropy (CE) [24, 57, 90, 110] quantify the degradation in predictive performance when salient features are masked or shuffled, while Relative Standard Deviation (RSD) [11] measures the variability in output confidence across perturbed samples. The Attention Allocation Measure (AAM) [53] evaluates the alignment between attention weights and semantically meaningful regions, and the AUC-Difference [30, 119] captures the reduction in discriminative ability caused by explanation-guided feature removal. Cumulative Prediction Difference (CPD) and Cumulative Prediction Preservation (CPP) [57, 68] track how model outputs change as salient features are progressively removed, revealing the model’s dependency on the identified regions. Their area-based extensions [68], including the Area Under Prediction Difference (AUPD) and Area Under Prediction Preservation (AUPP), integrate these effects over multiple k -values for smoother and more stable estimates. AUAUCD, AUAPRD, AUAUCP, and AUAPRP [68] summarize performance differences in terms of AUC or APR when salient features are perturbed. Sufficiency (Suff) [57, 90] quantifies how much the predicted probability changes when only the highlighted features are retained. Comprehensiveness (Comp) [57, 90] measures how much the prediction confidence decreases when the highlighted features are removed. Faithfulness@ k [23] evaluates the consistency of explanation quality over the top- k ranked features. AUROC Drop [149] captures the decline in performance after removing salient features, in which larger drops imply stronger causal importance.

Complexity and Smoothness. Complexity and Smoothness [12] quantify the sparsity of salient regions and the continuity of their transitions, respectively. Information and Mask Entropy [24, 33, 90] capture uncertainty within learned masks, where lower entropy implies clearer attributions. The Energy Retention Ratio (ERR) [160] measures the proportion of signal energy preserved in selected regions, while Smallest Deletion Features (SDF) [102] determines the minimal number of features that must be removed to produce a significant change in model output.

Robustness. Robustness metrics evaluate the stability of explanations under stochastic perturbations. Relative Output Stability (ROS) [12] quantifies the normalized change in the saliency map relative to the corresponding change in model output when Gaussian noise is added to the input. Lower ROS values indicate that explanations remain consistent under minor perturbations, reflecting higher robustness and reproducibility.

3.2 What Temporal Patterns Matter: Pattern-space Level

Signal-space explanations reveal which parts of the raw temporal signal contribute most to the model’s decision. However, their focus on isolated time points limits interpretive coherence, since such localized relevance seldom corresponds to how humans perceive temporal phenomena. Human reasoning typically integrates nearby observations into unified patterns that convey context and continuity. Building on this cognitive motivation, pattern-space explanations focus on identifying coherent temporal structures that extend beyond individual points. Instead of emphasizing isolated timesteps,

Table 3. Key highlights of pattern-space explanations. Abbreviations: SDD = Describe-and-Detect, RAG = Retrieval-augmented Generation, LCE = Local Cascade Ensemble.

Model	Explanation Type	Methodology	Task	Ante-hoc/Post-hoc
InteDisUX [150]	Temporal Segment	Integrated Gradients	Classification	Post-hoc
Implet [101]	Temporal Segment	Subsequence Extraction	Classification	Post-hoc
InterpGN [166]	Temporal Segment	Shapelet Bottleneck	Classification	Ante-hoc
SHAPEX [54]	Temporal Segment	SDD	Classification	Post-hoc
TS-RAG [113]	Temporal Segment	RAG	Forecasting	Ante-hoc
TIMEVIEW [63]	Temporal Segment	Composition Mapping	Forecasting	Ante-hoc
XEM [36]	Temporal Segment	LCE	Classification	Ante-hoc
ADSN [96]	Temporal Segment	Adversarial Training	Classification	Ante-hoc
Karlsson2020 [65]	Temporal Segment	Shapelet Tweaking	Classification	Post-hoc
ELIS [35]	Temporal Segment	Coverage Mechanism	Classification	Ante-hoc
LTS [41]	Temporal Segment	Shapelet Optimization	Classification	Ante-hoc
Shapelet Transform [87]	Temporal Segment	Search-based	Classification	Ante-hoc
DPSN [146]	Temporal Segment	Search-based	Classification	Ante-hoc
Logical Shapelets [108]	Temporal Segment	Search-based	Classification	Ante-hoc
TS Shapelets [172]	Temporal Segment	Search-based	Classification	Ante-hoc
LIMESegment [140]	Temporal Segment	Perturbation	Classification	Post-hoc
LEFTIST [47]	Temporal Segment	Perturbation	Classification	Post-hoc
ProtoTS [122]	Prototype	Hierarchical Learning	Forecasting	Ante-hoc
PTAP [21]	Prototype	Activation Extraction	Classification	Post-hoc
Ghosal2021 [40]	Prototype	Encoder-based	Classification	Ante-hoc
TapNet [178]	Prototype	Attention-based	Classification	Ante-hoc
Basisformer [112]	Prototype	Contrastive Learning	Forecasting	Ante-hoc
N-BEATS [115]	Prototype	Decomposition	Forecasting	Ante-hoc
Mr-SEQ [74]	Symbolic Pattern	Symbolic Sequence Learning	Classification	Ante-hoc
SAX-VSM [133]	Symbolic Pattern	TF-IDF	Classification	Ante-hoc
Srinivasan2025 [143]	Counterfactual	Gradient Optimization	Anomaly Detection	Post-hoc
Glacier [164]	Counterfactual	Gradient Optimization	Classification	Post-hoc
Sulem2022 [144]	Counterfactual	Gradient Optimization	Anomaly Detection	Post-hoc
LatentCF++ [165]	Counterfactual	Gradient Optimization	Classification	Post-hoc
Sub-SpaCE [128]	Counterfactual	Genetic Optimization	Classification	Post-hoc
TSEvo [52]	Counterfactual	Genetic Optimization	Classification	Post-hoc
CounTS [168]	Counterfactual	Variational Inference	Classification	Ante-hoc
CELS [80]	Counterfactual	Replacement	Classification	Post-hoc
SETS [10]	Counterfactual	Replacement	Classification	Post-hoc
MG-CF [81]	Counterfactual	Replacement	Classification	Post-hoc
Native Guide [28]	Counterfactual	Replacement	Classification	Post-hoc
CoMTE [9]	Counterfactual	Replacement	Classification	Post-hoc

this level characterizes how salient subsequences, motifs, or trends influence the model’s decision. By highlighting recurring temporal patterns within the signal, these explanations clarify which meaningful dynamics are most relevant to the prediction. Table 3 summarizes representative pattern-space methods.

3.2.1 Explanation Types. We organize the existing explanation methods into four major categories: temporal segments, prototypes, symbolic patterns, and counterfactual time series. These explanations are shown in Figure 5.

Temporal Segment. As shown in Fig. 5(b)(i), segment-based explanations highlight contiguous subsequences that drive predictions. Search-based shapelet methods identify discriminative subsequences via information gain or statistical tests. For example, Time Series Shapelets [172] discovers discriminative subsequences through search, selecting the most informative local patterns by information gain; through logical combination of discovered shapelets, Logical Shapelets [108] form expressive explanations from class-discriminative subsequences; Shapelet Transform [87] selects shapelets using information gain or F-statistic to capture class-distinguishing temporal patterns; by leveraging symbolic Fourier approximation (SFA) [131], DPSN [146] learns prototype-informed metrics and shapelet-like representations to explain few-shot classification.

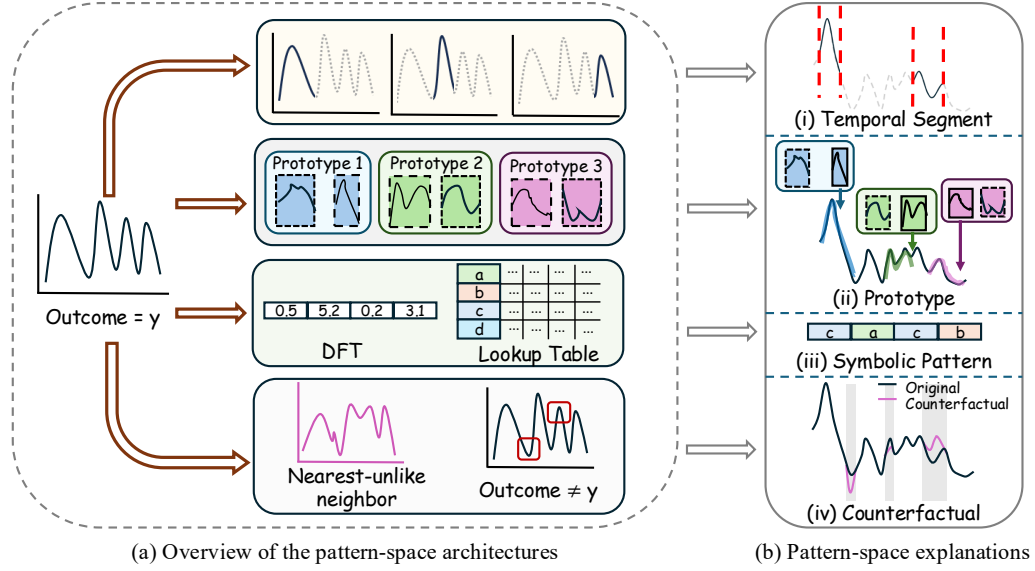


Fig. 5. Illustration of pattern-space methods. (a) is an overview of pattern-space architectures. These methods find useful time patterns by segmenting the input, matching it to prototypes, changing it into symbolic forms, or retrieving counterfactual examples. (b) shows the types of pattern-space explanations. Abbreviation: DFT = Discrete Fourier Transformation.

In contrast, learned shapelet methods learn discriminative subsequences by optimizing differentiable objectives or refining candidates: LTS [41] learns shapelets end-to-end by minimizing a differentiable classification objective with a soft-min distance; ELIS [35] mines candidates in the Piecewise Aggregate Approximation (PAA) space [161], selects them by coverage, and refines with logistic regression; Karlsson et al. [65] explain decisions by detecting and minimally tweaking discriminative subsequences that flip predictions; through adversarial training, ADSN [96] produces sample-specific dynamic shapelets that resemble real subsequences; SHAPEX [54] aligns time series to shapelet-defined segments via Shapelet Describe-and-Detect (SDD), then applies segment-level perturbations with Shapley-value attribution to produce faithful segment explanations; InterpGN [166] combines a shapelet-bottleneck model with a deep expert and uses a confidence-based gate to choose the expert for each instance; Implet [101] extracts post-hoc subsequence evidence by selecting continuous high-attribution segments from saliency maps.

Moreover, perturbation-based segment methods estimate segment contributions by perturbing fragments and measuring prediction changes: LEFTIST [47] uses segment-level perturbations with a local linear surrogate to estimate each segment's contribution to the prediction; by generating frequency-domain perturbations on homogeneous signal partitions, LIMESegment [140] trains a local linear surrogate to assess segment-level contributions. In addition, XEM [36] leverages a Local Cascade Ensemble (LCE) framework that explains its predictions by pinpointing the most discriminative temporal subsequence. By using composition mapping based on B-spline basis [27] decomposition, TIMEVIEW [63] provide ante-hoc explanation by decomposing predicted trajectories into interpretable temporal segments and their underlying dynamical motifs. InteDisUX [150] computes segment-level importance via Integrated Gradients (IG) [145], then iteratively merges adjacent segments guided by discrimination and faithfulness gains to produce explanations.

TS-RAG [113] explains forecasts by retrieving top- k analogue time series segments from a knowledge base and using a learnable adaptive retrieval mixer to weight and fuse them as evidence.

Prototype. Prototype-based explanations attribute predictions to representative subsequences in a learned latent space, as illustrated in Fig. 5(b)(ii). Salient examples include TapNet [178], which encodes time series with LSTM–CNN layers, applies random dimension permutation and learns attention-weighted class prototypes in latent space. Based on a maximum-mean discrepancy (MMD)-based selection criterion [67], PTAP [21] identifies highly activated temporal patterns and chooses representative prototypes. Ghosal et al. [40] propose a modular prototype-learning framework that extracts single-variable prototypes and aggregates their similarities to form multivariable prototypes. By computing instance–prototype similarities on denoised representations, ProtoTS [122] organizes prototypes hierarchically to provide multi-level explanations. A closely related line learns global bases that act as reusable prototypical: N-BEATS [115] decomposes time series into explainable trend and seasonality bases via a neural basis expansion with forward–backward residual blocks; adopting a self-supervised contrastive learning objective, Basisformer [112] learns explainable bases that align relationships between time series and bases across historical and future views.

Symbolic Pattern. Symbolic pattern explanations (Fig. 5(b)(iii)) represent time series as sequences of discrete symbols and attribute predictions to human-readable patterns such as words and logical compositions. Based on symbolic aggregation (SAX) [120], SAX-VSM [133] transforms time series into symbolic words, builds class-level TF–IDF vectors, and ranks the most informative words as explanation patterns. Mr-SEQ [74] learns weighted symbolic subsequences from multi-resolution, multi-domain representations using linear models, and maps these features back to the original signal to explain predictions.

Counterfactual Time Series. Replacement-based counterfactuals replace contiguous input segments with plausible alternatives from nearby examples or interpretable primitives. For example, CoMTE [9] generates counterfactual explanations through variable substitution from a similar instance, using greedy or hill-climbing search to find minimal changes that alter the prediction. Native Guide [28] produces counterfactual explanations by adapting nearest unlike neighbors (NUNs) using saliency-guided subsequence perturbations. By replacing motif segments with representative motifs from the target class, MG-CF [81] generates counterfactual explanations. SETS [10] uses shapelets as explainable building blocks to generate counterfactual time series explanations, indicating which local temporal patterns must change to alter the model’s prediction. CELS [80] learns saliency maps and performs NUNs replacements to obtain interpretable counterfactuals. Genetic optimization counterfactuals use evolutionary search over segment edits to find sparse, plausible changes that flip the prediction while staying close to the original. LatentCF++ [165] generates counterfactual explanations by applying gradient-based perturbations in the latent space of an auto-encoder to modify time series representations. Sulem et al. [144] generate explanations through multi-objective counterfactual optimization in a latent space. Leveraging gradient-based optimization guided by temporal constraints, Glacier [164] generates counterfactual explanations that remain realistic and focused. Srinivasan et al. [143] generate explanations through gradient-based counterfactual optimization combined with feature selection. On the other hand, genetic optimization counterfactuals search the space of segment edits with evolutionary objectives that balance flip, proximity, plausibility, and sparsity. TSEvo [52] applies evolutionary perturbations that iteratively adjust temporal segments to create realistic and class-changing instances. Sub-SpaCE [128] utilizes a multi-objective genetic optimization framework that enforces sparsity, plausibility, and contiguity to derive counterfactual explanations. Additionally, with a variational Bayesian causal model, CountTS [168] performs counterfactual inference via abduction–action–prediction reasoning.

3.2.2 Evaluation. The evaluation of pattern-space explanations aims to assess whether the identified temporal patterns are faithful to the model’s underlying prediction. Unlike signal-space evaluation, which focuses on point-wise saliency accuracy, pattern-space evaluation emphasizes the consistency, stability, and realism of higher-level temporal abstractions such as shapelets, prototypes, or counterfactual trajectories. Quantitative methods cover *faithfulness*, *validity*, *proximity*, *sparsity*, *plausibility*, *diversity*, and *robustness*.

Qualitative Evaluation. At the pattern-space level, qualitative evaluation focuses on assessing whether the visualized temporal patterns are consistent with domain knowledge. Such evaluation is less common and typically takes the form of controlled user studies. This type of evaluation is illustrated in [47], where participants were asked to identify the model’s predicted class from highlighted time series segments. The resulting accuracy indicates how clearly the explanations convey discriminative temporal patterns.

Faithfulness. Faithfulness captures how strongly the identified or generated temporal patterns influence model predictions. It reflects the causal impact of highlighted regions on the model’s decision process. For segment explanations, Faithfulness Score [101, 140, 150] measures the degradation in model performance after removing or perturbing important subsequences. Another metric, Fidelity [47], focuses on the agreement between the surrogate local model and the opaque classifier. In counterfactual explanations, the Counterfactual Accuracy, the Mean Squared Error (MSE), and the Counterfactual Change Ratio (CCR) [168] metrics quantify how effectively the generated examples achieve the desired label transitions, maintain prediction stability, and confine modifications to label-relevant regions.

Validity. While faithfulness evaluates whether explanations align with model reasoning, validity ensures that generated counterfactuals achieve the intended outcome change. The Label Flip Rate [52, 128, 143, 164, 165] quantifies the proportion of counterfactuals that successfully cross the decision boundary to the target class. Conversely, the Failure Rate [144] provides an inverse view by measuring the proportion of invalid counterfactuals. A lower value indicates that the method can produce transformations that adhere to the task’s rules. Similarly, the Weighted Adjusted Rand Index (WARI) [14] evaluates the validity of time series segmentation — it uses weights based on when an error occurs, meaning that deviations close to boundaries are treated more leniently while errors in regions intended to remain stable get stronger penalties.

Proximity. Proximity measures the distance between the counterfactual and the original series, ensuring that perturbations remain small enough to be understandable. Several norms are commonly used for this purpose. The L_1 -norm [80, 81] calculates the average absolute deviation between the counterfactual and the original series, while the L_2 -norm and L_∞ -norm [10, 128, 164, 165] quantify Euclidean and maximum deviations, respectively. The Relative Counterfactual Distance (RCF) [28] takes a different approach. It computes the distance between the counterfactual and the query instance compared to the distance to the nearest example from a different class. This ratio places proximity within the context of the surrounding data distribution. Dynamic Time Warping (DTW) [144] measures temporal alignment, emphasizing shape-level similarity between original and counterfactual trajectories. Smaller distances in these metrics correspond to more minimal and explainable counterfactuals.

Sparsity. Sparsity characterizes the minimality of change by quantifying the proportion of altered elements relative to the length of the time series. The Sparsity Level [10, 28, 80, 81, 128, 143] measures the proportion of elements that remain unchanged after perturbation, or equivalently, the fraction of altered features normalized by the input dimensionality, making it conceptually equivalent to the L_0 -norm [52]. The two differ only in normalization and interpretation, with higher normalized values indicating greater sparsity.

Table 4. Key highlights of structural explanations. Abbreviations: SLS = Stable Latent Structure, RRP = Regression Relevance Propagation, GC = Granger Causality.

Model	Explanation Type	Methodology	Task	Ante-hoc/Post-hoc
Mekonnen2024 [100]	Rule-based	Decision Tree	Classification	Post-hoc
Aasi2023 [1]	Rule-based	Decision Tree	Classification	Ante-hoc
LASTS [46]	Rule-based	Decision Tree	Classification	Post-hoc
TemporalRule [163]	Rule-based	Neuro-symbolic Learning	Classification	Ante-hoc
Pagliarini2024 [116]	Rule-based	Neuro-symbolic Learning	Classification	Ante-hoc
NSTSC [170]	Rule-based	Neuro-symbolic Learning	Classification	Ante-hoc
STONE [169]	Rule-based	Neuro-symbolic Learning	Classification	Ante-hoc
NN-TLI [78]	Rule-based	Temporal Logic Inference	Classification	Ante-hoc
Sara2020 [106]	Rule-based	Enumeration-based Search	Classification	Post-hoc
RCNs [114]	Rule-based	Rule-constrained Network	Classification	Ante-hoc
PHAR [107]	Rule-based	Numeric-to-rule Transformation	Classification	Post-hoc
LIMREF [125]	Rule-based	Rule Mining	Forecasting	Post-hoc
CAP [29]	Rule-based	Rule Mining	Forecasting	Post-hoc
OracleAD [20]	Causal Graph	SLS Alignment	Anomaly Detection	Ante-hoc
CausalFormer [69]	Causal Graph	RRP	Causal Discovery	Ante-hoc
CUTS [19]	Causal Graph	Variational Causal Structure Learning	Causal Discovery	Ante-hoc
ACD [91]	Causal Graph	Amortized Encoder	Causal Discovery	Ante-hoc
GVAR [99]	Causal Graph	Generalised Vector Autoregression	Causal Discovery	Ante-hoc
TCCL [55]	Causal Graph	Nonlinear GC learning	Classification	Ante-hoc
DFCM [158]	Causal Graph	Fuzzy Logic	Forecasting	Ante-hoc
TCDF [111]	Causal Graph	Attention	Forecasting	Ante-hoc
Z-Time [76]	Event Relation	Temporal Relation Mining	Classification	Ante-hoc

Plausibility. Plausibility examines the realism of generated patterns, ensuring that counterfactuals stay within the data manifold. On the other hand, Local Outlier Factor (LOF), Isolation Forest (IF), and One-Class SVM (OC-SVM) [10, 28] assign lower outlier scores to counterfactuals that better conform to the training distribution. The Increment in Outlier Score (IOS) [128] metric evaluates distributional deviation using autoencoder reconstruction errors, where values closer to zero indicate higher plausibility. The neighborhood-consistency metric yNN [52] evaluates local data support by measuring the proportion of nearby training samples sharing the same class label as the counterfactual’s. Implausibility indices [144] capture complementary aspects of realism, including DTW distance to the median forecast (Implausibility 1), temporal smoothness (Implausibility 2), and negative log-likelihood under the predictive distribution (Implausibility 3). Lower scores across these metrics indicate counterfactuals that are more coherent and realistic. Beyond these, the State Matching Score (SMS) [14] metric evaluates the plausibility of time series segmentation by categorizing errors into delay, isolation, transition, and missing types, and weighting them according to their semantic severity.

Diversity. Diversity quantifies the variability within a counterfactual ensemble by measuring the variance of counterfactual examples across time steps [144]. Here, higher variance indicates broader temporal coverage and richer diversity within the target distribution.

Robustness. These metrics evaluate the stability of explanations under perturbations or sampling variations. For example, Robustness Score [140] quantifies the proportion of time series whose explanations remain unchanged after adding random noise, with higher values indicating more stable interpretations. Additionally, the Local Lipschitz Constant [9] can be used to capture sensitivities by computing the ratio between changes in explanations and corresponding input perturbations across local neighborhoods, in which lower values denote smoother and more robust explanations.

3.3 How is the Decision Made: Structural Level

Pattern-space explanations capture representative subsequences that summarize temporal regularities. However, they often overlook the dependencies among patterns. This limitation motivates the development of structural level methods,

which advance from pattern recognition to relational reasoning. By uncovering how temporal components interact through rules, causal links, or event relations, this level aligns more closely with how humans construct explanatory reasoning rather than merely identify repeated shapes. Structural explanations capture model reasoning through human-readable representations such as rules, graphs, or causal structures. The goal at this level is to reveal how the model organizes dependencies among temporal or feature components to reach a decision. From this structural perspective, explanations illustrate how different relationships jointly contribute to the model’s prediction. Table 4 lists representative structural explanation methods.

3.3.1 Explanation Types. We group structural-level explanations into three types: rule-based, causal structure, and event-based. Figure 6 offers a visual summary of the different types.

Rule-based. As shown in Fig. 6(b)(i), rule-based explanations express decisions as readable rules learned from temporal features or temporal logic. Rule-Constrained Networks (RCNs) [114] constrain neural networks to select decisions from a given rule set so predictions are supported by explicit rules while maintaining high accuracy. The decision tree methods fit interpretable trees to learned or extracted temporal features and output human-readable rules: LASTS [46] learns shapelet-aware rule explanations by training an interpretable decision tree on exemplar and counter-exemplar sets extracted around the instance; by incrementally learning Signal Temporal Logic (STL) [98] rules through decision trees, Aasi et al. [1] propose a time-incremental learning framework for explainable rule induction; extracting parameterized event primitives from time series, Mekonnen et al. [100] cluster explainable features and train a decision-tree surrogate that mimics deep model predictions to yield global temporal rules. In addition, combining neural models with temporal-logic formalisms, neuro-symbolic methods learn symbolic rules that explain decisions: Through formula enumeration and pruning using formula signatures, Mohammadinejad et al. [106] identify concise Parametric Signal Temporal Logic (PSTL) [6] rules; NN-TLI [78] encodes STL operators with differentiable time functions and sparse softmax, enabling direct learning of compact, interpretable temporal-logic formulas; STONE [169] integrates neural networks with weighted Signal Temporal Logic (wSTL) to learn symbolic temporal rules; grounded in neuro-symbolic reasoning, NSTSC [170] extracts wSTL formulas within a decision-tree based framework to generate rules; Pagliarini et al. [116] express model decisions as temporal-logic rules, enabling symbolic reasoning over intervals while retaining neural representational power; built upon STL operator embedding, TemporalRule [163] learns symbolic temporal rules that are consistent with the model’s predictions. Furthermore, rule mining builds rule sets from neighborhoods, case-control analyses, or post-hoc attributions: CAP [29] integrates the case-crossover design with the Apriori algorithm [56] to extract causal patterns by comparing control and case periods within time series; LIMREF [125] generates local rule-based explanations by mining impact rules from synthetic neighborhoods, producing affirmative and counterfactual conditions that show how feature settings affect forecasts; by converting numeric attributions into symbolic rules and fusing them across explainers, PHAR [107] derives temporal rules that summarize the most influential time-feature intervals and resolve conflicting attributions into a high-confidence explanation for each instance.

Causal Structure. Causal-structure explanations focus on estimating temporal causal links and presenting them as explainable networks (Fig. 6(b)(ii)). TCDF [111] uses an attention-based convolutional network to infer causal links and delays through attention interpretation and permutation validation; with a residual graph learning framework, TCCL [55] learns class-specific nonlinear Granger-causal graphs within a recurrent architecture to handle data imbalance; DFCM [158] integrates deep neural networks into fuzzy cognitive maps [71]; ACD [91] leverages an amortized encoder-decoder framework to infer causal graphs, with the predicted edges providing interpretable Granger-causal explanations; using alternating optimization, CUTS [19] jointly imputes missing values in irregularly observed time

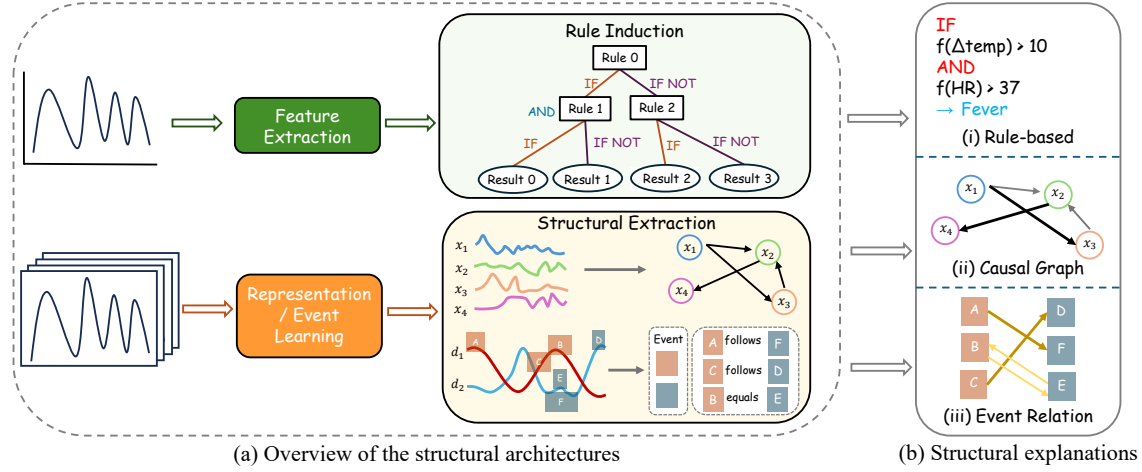


Fig. 6. Illustration of structural level methods. (a) an overview of structural-level architectures. These methods provide explanations by creating rules or learning representations that capture event patterns and time relationships. (b) types of structural level explanations.

series and estimates sparse nonlinear Granger-causal adjacency matrices; CausalFormer [69] discovers temporal causal graphs by decomposing a causality-aware transformer using Regression Relevance Propagation (RRP). Finally, built upon temporal causality modeling and Stable Latent Structure (SLS) alignment, OracleAD [20] explains anomalies by diagnosing root causes through dual scores reflecting prediction and structural deviation.

Event-based. As illustrated in Fig. 6(b)(iii), Z-Time [76] abstracts raw signals into event intervals, mines temporal relations among them, and uses these symbolic patterns to build an interpretable multivariate classifier.

3.3.2 Evaluation. At the structural level, evaluation examines whether the extracted symbolic rules, causal graphs, or event dependencies accurately capture the model’s reasoning process and can be assessed by domain experts for validity. Because structural explanations make reasoning explicit through formal logic or causal structures, their assessment further involves quantitative evaluation, measuring *fidelity*, *consistency with the model’s predictive behavior*, *Confidence*, *Robustness*, *Usefulness*, and *complexity*.

Qualitative Evaluation. Qualitative evaluation focuses on how symbolic rules, temporal formulas, and causal graphs convey a model’s reasoning in a interpretable form. It emphasizes expert-based assessment, examining whether the extracted structures correspond to semantically meaningful temporal or causal mechanisms. Domain experts are invited to interpret the learned rules or causal dependencies and to evaluate their logical soundness, physical plausibility, and consistency with domain knowledge [29, 55]. For the causal-graph model TCCL [55], aviation engineers qualitatively inspect the learned nonlinear Granger-causal graphs to assess whether the directed dependencies among variables reflect plausible thermodynamic or mechanical relationships in aircraft engines. Similarly, in the rule-based causal framework CAP [29], industrial process experts review the extracted association and causal rules explaining flooding events in petroleum distillation, verifying whether the identified precursors and logical conditions align with known process dynamics and safety mechanisms. Such expert feedback provides external validation that the explanations are scientifically coherent and domain-consistent, ensuring that the model’s learned logic aligns with real-world mechanisms rather than remaining confined to its internal computation.

Table 5. Key highlights of semantic explanations. Abbreviations: SM = Summary, SR = Stepwise Reasoning, RG = Retrieval-Grounded Text, EG = Example-Grounded Text, and RB = Rule-Based Text, CoT = Chain-of-thought, RAG = Retrieval-augmented Generation, C = Classification, F = Forecasting, I = Imputation, A = Anomaly Detection, O = Open-Ended Reasoning, QA = Question Answering.

Model	Explanation Type	Methodology	Task	Ante-hoc/Post-hoc
Time-MQA [70]	SM	Direct Prompting	C/F/I/A/O/QA	Ante-hoc
Chat-TS [124]	SM	Direct Prompting	QA	Ante-hoc
ReDP [7]	SM	Direct Prompting	Forecasting	Ante-hoc
LLMTime_E [3, 42]	SM	Direct Prompting	Forecasting	Ante-hoc
JoLT [13]	SM	Direct Prompting	Summarization	Ante-hoc
FSTLLM [58]	EG	In-context Learning	Forecasting	Ante-hoc
AXIS [72]	EG	In-context Learning	Anomaly Detection	Ante-hoc
xTP-LLM [50]	EG	In-context Learning	Forecasting	Ante-hoc
TimeXL [61]	EG	In-context Learning	Classification	Ante-hoc
TimeCAP [75]	EG	In-context Learning	Classification	Ante-hoc
Yu2023 [174]	EG	In-context Learning	Forecasting	Ante-hoc
GEM [73]	SR	CoT Prompting	C/A	Ante-hoc
TimeMaster [177]	SR	CoT Prompting	Classification	Ante-hoc
Time-RA [171]	SR	CoT Prompting	Anomaly Detection	Ante-hoc
Time-R1 [94]	SR	CoT Prompting	Forecasting	Ante-hoc
TimeReasoner [157]	SR	CoT Prompting	Forecasting	Ante-hoc
STRATA-TS [59]	RG	RAG	Forecasting	Ante-hoc
ReasonTSC [181]	EG/SR	In-context/CoT	Classification	Ante-hoc
Dong2024 [31]	SM/EG/SR	Direct/In-context/CoT	Anomaly Detection	Ante-hoc
ZARA [82]	RG/SR	RAG/CoT	Classification	Ante-hoc
LLMAD [88]	RG/SR	RAG/CoT	Anomaly Detection	Ante-hoc
TSXplain [109]	RB	Natural Language Generation	Anomaly Detection	Post-hoc

Fidelity and Predictive Alignment. These metrics assess how effectively an extracted structure can represent the behavior of the original model. Fidelity, Accuracy, and Misclassification Rate (MCR) [46, 100, 106, 125] are all used to quantify how closely a surrogate explanation aligns with the opaque model’s predictive behavior. Fidelity measures alignment with model predictions, while Accuracy and MCR evaluate consistency with the ground-truth labels. For graph-level explanations, AUROC, AUPR, and Balanced Accuracy (BA) [55, 99] assess the discriminative reliability of the inferred causal graphs, where AUROC captures ranking consistency, AUPR emphasizes precision in positive-class detection, and BA considers both true positive and true negative performance.

Rule Confidence. Confidence and Support [116] are two basic measures used to quantify the statistical reliability and representativeness of symbolic rules. Confidence measures the conditional probability that a rule’s antecedent implies its consequent, while Support indicates its frequency within the dataset.

Robustness and Usefulness. Robustness [100] measures the stability of rules or causal relationships when the inputs are slightly perturbed. Usefulness, often measured via 1-NN classification accuracy [46], evaluates whether the symbolic rules maintain predictive effectiveness on unseen data instances.

Complexity and Coherence. Complexity and Incoherence [46] capture the redundancy and overlap among rules, with lower values indicating more compact and comprehensible explanations.

3.4 Why It Happens: Semantic Level

Structural explanations convey the internal logic of a model through symbolic rules or causal graphs, revealing how variables or events interact to produce outcomes. Despite their clarity, these representations remain formal and abstract, often requiring expert interpretation. The emergence of large language models (LLMs) has enabled a shift from symbolic reasoning to human-like communication. Their ability to encode conceptual knowledge and generate coherent language makes it possible to translate structural reasoning into narratives or concept-grounded expressions that people can

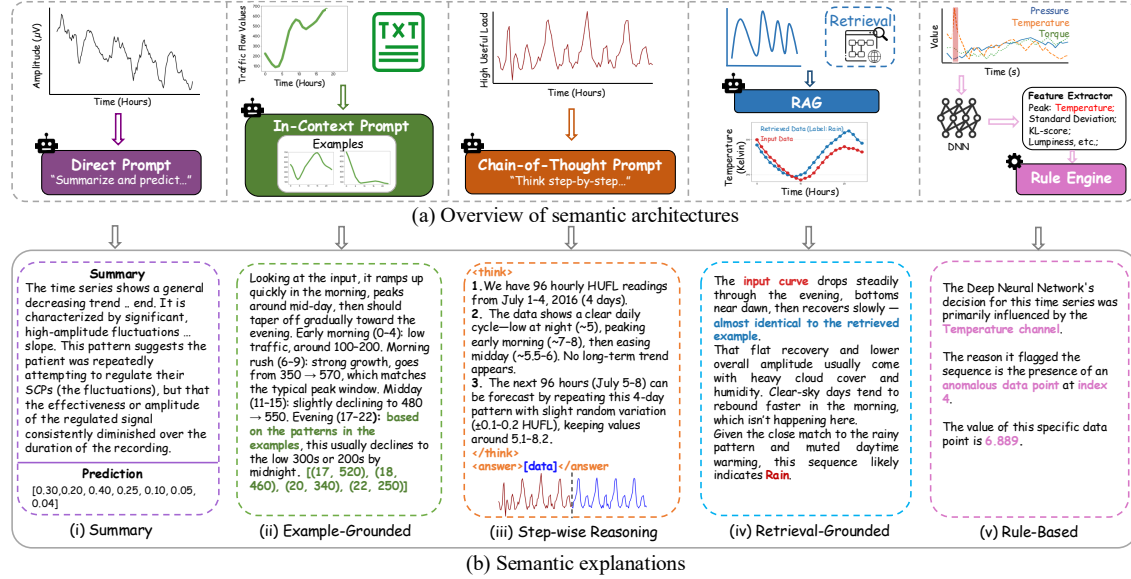


Fig. 7. Illustration of semantic-level explanation methods. (a) is an overview of semantic-level architectures. These methods use LLMs via direct queries, in-context examples, stepwise reasoning, and retrieval-augmented generation, or rely on rule-based engines. (b) shows the types of semantic-level explanations.

intuitively understand. At the semantic level, explainability reaches its highest degree of abstraction, where model reasoning is expressed through natural language or conceptual narratives. Explanations at this stage describe why specific outcomes occur by integrating instructions, contextual information, or domain knowledge into coherent textual descriptions. Such explanations bridge the gap between computational reasoning and human understanding, allowing models to communicate their decisions in a form that resembles human explanation. Table 5 presents representative methods for semantic explanations.

3.4.1 Explanation Types. At the semantic level, we classify textual explanations into five types: (1) summary, (2) example-grounded, (3) stepwise reasoning, (4) retrieval-grounded, and (5) rule-based text. The explanation types are shown in Figure 7.

Summary. As illustrated in Fig. 7(b)(i), summary explanations provide brief natural-language descriptions of a time series' dominant patterns to accompany the prediction. On that lens, JoLT [13] jointly learns aligned embeddings for time series and language through a multi-modal encoder-decoder framework, enabling textual descriptions that reflect temporal dynamics and serve as semantically grounded explanations. LLTime_E [42] explains time series forecasting by prompting a large language model (LLM) to identify the underlying generating function of a given numerical sequence. Then, it generates a textual analysis on the series' properties to explain its reasoning, to justify its selection. By adopting advanced prompting strategies such as chain-of-thought (CoT) and context-aided prompting, ReDP [7] elicits explicit reasoning traces that serve as human-readable explanations of model decisions. Chat-TS [124] generalizes to multi-modal reasoning by jointly modeling time series and textual data, where the LLM engages in dialogue-like reasoning to integrate linguistic context and generate human-like interpretive commentary. Formulated as

a multi-task question-answering problem, Time-MQA [70] also enables LLMs to answer structured temporal questions, offering interpretive insight into both predictions and causal relations.

Example-Grounded Text. As illustrated in Fig. 7(b)(ii), example-grounded explanations compare the input with relevant examples or signals and tell how each of them helps explain the result. Yu et al. [174] introduce an explainable financial forecasting framework where time series signals are converted into textual representations and analyzed through LLM-based reasoning, producing human-readable explanations that link temporal indicators with economic semantics and market events. By integrating structured spatio-temporal modeling with LLM reasoning, xTP-LLM [50] summarizes extracted traffic patterns into structured prompts that guide the LLM to generate natural-language narratives relating forecast results to causal factors, such as weather and congestion. Within an in-context surrogate framework, TimeCAP [75] uses LLM agents to contextualize, augment, and predict event sequences, producing dialogue-style rationales that capture temporal causality and external knowledge. TimeXL [61] integrates LLMs into a multi-modal forecasting pipeline, where textual reasoning complements time series modality. The LLM interprets temporal features and generates explanatory narratives describing causal and contextual relationships that underlie predictions. FSTLLM [58] presents a spatio-temporal LLM for few-shot forecasting that explicitly verbalizes spatial and temporal dependencies. AXIS [72] employs an LLM-based anomaly detection framework that transforms inputs into structured prompts with hints, allowing the LLM to generate cause-oriented narratives that explain anomaly types, causes, and severities with end-to-end explainability.

Stepwise Reasoning. As illustrated in Fig. 7(b)(iii), stepwise reasoning explanations show the decision process step by step, then present the final prediction. To achieve this, by incorporating feedback loops between the LLM and the anomaly detector, Time-RA [171] refines explanations during dataset annotation, training a model for causal reasoning about temporal irregularities. TimeReasoner [157] conceptualizes temporal prediction as a reasoning task, where LLMs emulate deliberative thought to articulate intermediate reasoning steps before producing forecasts. Using a two-stage reinforcement fine-tuning framework that combines supervised CoT learning with GRIP-based policy optimization, Time-R1 [94] enables LLMs to perform slow-thinking reasoning, generating structured reasoning traces and textual forecasts with enhanced accuracy and explainability. Similarly, TimeMaster [177] advances LLM-based time series reasoning by incorporating reinforcement learning and multi-modal grounding, producing rich textual rationales that explain the decision process across modalities. Finally, to address multi-modal grounding, GEM [73] links ECG series and medical images through a vision-language model that produces textual explanations, clarifying physiological meaning behind abnormal signals.

Retrieval-Grounded Text. As illustrated in Fig. 7(b)(iv), through selective knowledge retrieval for urban time series forecasting, STRATA-TS [59] also integrates domain knowledge into reasoning chains, producing explanations that connect environmental factors with observed dynamics.

Rule-Based Text. As illustrated in Fig. 7(b)(v), TSXplain [109] combines influence tracing with statistical feature extraction to translate salient patterns into natural-language explanations via rule templates, and using sanity checks to ensure consistency with the model’s decision.

Multiple Types. Existing works also integrate multiple types for improving interpretation. For example, by applying direct, in-context, and chain-of-thought (CoT) prompting to textualized temporal segments, Dong et al. [31] enable LLMs to generate concise natural-language justifications clarifying why anomalies occur. ReasonTSC [181] enhances LLM reasoning by combining in-context demonstrations with CoT prompting to elicit structured, stepwise reasoning,

producing interpretable decision paths that explain class assignments. LLMA [88] detects anomalies via retrieval-enhanced in-context learning and an Anomaly Detection Chain-of-Thought (AnoCoT) that injects domain rules and expert reasoning, enabling LLMs to generate human-readable explanations specifying anomaly types, severity, and causes for each detected event. Inspired by retrieval and knowledge-driven agentic design, ZARA [82] combines external knowledge bases with LLM reasoning to produce both causal and semantic explanations for motion anomalies.

3.4.2 Evaluation. At the semantic level, evaluation seeks to measure how well text explanations convey the reasoning, evidence, and decision process of time series models in a form understandable to humans. Semantic explanations are expressed through free-form text, which makes their evaluation fundamentally different from traditional attribution metrics, and is primarily based on qualitative assessment, with quantitative measures such as *simulatability accuracy* and *reference-based metrics* applied in limited cases.

Qualitative Evaluation. Semantic-level explanations are abstracted in natural language, making evaluations rely on experts or LLM-as-a-judge approaches to assess the quality of the generated interpretations under multiple criteria:

- **Humans-as-judges.** A common strategy is to solicit domain experts to score generated explanations. These scores typically assess dimensions such as *accuracy*, *completeness*, *clarity*, *usefulness*, and *readability*. For example, experts have been previously employed to assess explanations in time series question answering [70] and anomaly detection reasoning [88], in which outputs are graded across multiple semantic criteria. Although these evaluations are able to provide subjective assessments of model performance, they can be labor intensive and are subject to being noisy.
- **LLMs-as-judges.** To reduce the burden of human labor, LLMs are employed as tools to assess semantic explanations, providing a set of scores for each requirement fulfilled in the interpretation. These scores capture dimensions such as *evidence-based reasoning*, *relevance*, *completeness*, *accuracy*, *specificity*, *appropriateness*, and *depth*. For example, GPT-4o has been used to assign scores reflecting the clinical diagnostic fidelity and reasoning quality of ECG explanations [73]. Similarly, Gemini 2.5 is incorporated to evaluate multi-format question explanations using metrics specific to each question type: for multiple-choice questions, it assigns scores for *correctness* and *reasoning quality*; for open-ended questions, it uses *relevance*, *completeness*, and *accuracy*; and for true/false questions, it considers *correctness* along with *justification quality* [72]. Finally, LLM-generated scores have been used to reward improvements in chain-of-thought extensions according to heuristics such as *specificity*, *appropriateness*, *relevance*, and *depth*, which can then be combined with hard formatting signals and optimized under RL-like objectives [177]. Such metrics provide more scalable and reproducible evaluations of explanation quality, complementing human judgments.

Direct Simulatability Accuracy. This metric quantifies how effectively an explanation can convey the model’s prediction to a surrogate reasoner, such as a human or LLM [3]. Specifically, given a time series input H , a model produces a forecast $F = FM(H)$, and an explainer generates a natural language explanation $NLE = \text{Explainer}(H, F)$. A surrogate reasoner then uses NLE and H to produce an estimated forecast F' . Thus, accuracy is measured as the similarity between F and F' , for example using metrics like sMAPE, capturing whether the explanation faithfully communicates the model’s reasoning and predictive outcome. Here, we assume there is no data breach from the forecast, F , into the explanation, NLE .

Reference-based Metrics. Generated explanations are frequently evaluated by how closely they align with predefined gold-standard references. [13] uses metrics such as ROUGE-1, ROUGE-2, and ROUGE-L to quantify varying ranges of n-gram overlap and longest common subsequences, typically reporting Precision, Recall, and F1 scores. To capture semantic nuances, METEOR is then used to extend these measurements by accounting for synonymy, stemming, and

Table 6. Summary of synthetic datasets. Abbreviations: UTS = Univariate Time Series, MTS = Multivariate Time Series, C = Classification, R = Regression, A = Anomaly Detection, CD = Causal Discovery.

Data	Data Type	Task	Explanation Level
WebTraffic	UTS	C	Signal-space [32]
FreqShapes	UTS	C	Signal-space [89, 123, 175]
SeqComb-UV	UTS	C	Signal-space [89, 123, 175]
Synthetic Traces	UTS	C	Structural [106]
Switch-Feature	MTS	C	Signal-space [57, 90, 149, 151]
State	MTS	C	Signal-space [24, 57, 90, 149, 151]
Spike	MTS	C	Signal-space [77, 149]
Toy	MTS	C	Pattern-space [168]
SeqComb-MV	MTS	C	Signal-space [89, 123, 175]
LowVar	MTS	C	Signal-space [89, 123, 175]
Naval Surveillance	MTS	C	Structural [1, 78]
Urban Driving	MTS	C	Structural [1]
Simulated Full Flight	MTS	C	Structural [55]
Rare-Time	MTS	R	Signal-space [24, 90]
Rare-feature	MTS	R	Signal-space [24]
Rare-Observation	MTS	R	Signal-space [90]
Semantic Benchmark	UTS	A	Semantic [72]
Machine Benchmark	UTS	A	Semantic [109]
SKAB	MTS	A	Pattern-space [143]
Lorenz 96 model	MTS	CD	Structural [19, 69, 99]
fMRI	MTS	CD	Structural [19, 69, 99, 111]
Netsim	MTS	CD	Structural [19, 91]

paraphrasing, thereby granting partial credit for semantically equivalent matches. Subsequently, a learned metric BLEURT assesses alignment with human preference by measuring fluency, grammatical correctness, and semantic adequacy. Together, these metrics provide numerical signals regarding how well natural-language explanations approach the content and style expected by human evaluators.

3.5 Discussion

The four levels of explanation discussed in Section 3.1-3.4 (Figure 3) form a cognitive continuum that reflects the human process of understanding temporal information. Each level contributes a distinct perspective: *signal-space* explanations capture perceptual attention, *pattern-space* explanations extract temporal structures, *structural reasoning* reveals the underlying relational logic, and *semantic explanations* translate this logic into conceptual and communicative forms. This integrated hierarchy highlights the evolution of explainability from model transparency to human understanding.

4 Datasets and Applications

This section provides an overview of commonly used datasets and their applications. It shows how explainability research covers different domains.

4.1 Datasets

To provide a clear overview of the data discussed, Table 6, Table 7, and Table 8 summarize datasets used for all models in this survey. These tables include both synthetic and real-world scenarios, each showing the data’s domain, type, related time series task, explanation level, and source.

Synthetic Datasets. Table 6 lists the synthetic time series datasets commonly used for evaluating explanation methods. These datasets create controlled environments, which makes them suitable for evaluating signal-space and pattern-space

Table 7. Summary of real-world datasets. Abbreviations: UTS = Univariate Time Series, MTS = Multivariate Time Series, Multi = Multi-modal Time Series, C = Classification, F = Forecasting, A = Anomaly Detection.

	Data	Data Type	Task	Explanation Level
Healthcare	MIMIC-III	MTS	C	Signal-space [57, 68, 77, 90]; Pattern-space [140, 166]
	PhysioNet 2019	MTS	C	Signal-space [68]
	EHR	MTS	C	Signal-space [22]
	Cough&Breath Audio	MTS	C	Structural [116]
	Respiratory Sound	MTS	C	Structural [116]
	SHHS	MTS	C	Pattern-space [168]
	MESA	MTS	C	Pattern-space [168]
	SOF	MTS	C	Pattern-space [168]
	Test-Positive	Multi	C	Semantic [61, 75]
	Mortality	Multi	C	Semantic [61, 75]
	VitalDB	MTS	F	Semantic [157]
	Illness	MTS	F	Pattern-space [112]
	Mammography	UTS	A	Semantic [109]
Environment	ECG-Bench	Multi	A	Semantic [73]
	PTB-XL	Multi	QA	Semantic [13]
	EAM	UTS	C	Structural [106]
	Electricity	MTS	C	Signal-space [39, 85, 117, 156]; Pattern-space [112]
	Wound Healing	MTS	C	Structural [170]
	Solar Flare	MTS	C	Pattern-space [10]
	New York	Multi	C	Semantic [61, 75]
	PM2.5	MTS	F	Signal-space [39, 49, 53, 156]
	Air Quality	MTS	F	Signal-space [118]; Structural [158]
	Boiler	MTS	F	Signal-space [57, 123, 175]
	Energy	MTS	F	Signal-space [39]
	FUZZ-IEEE	MTS	F	Structural [125]
	Temperatures	MTS	F	Structural [158]
Sensor	AQWan&AQShunyi	MTS	F	Semantic [94]
	Wind	MTS	F	Semantic [94, 157]
	ECL	MTS	F	Semantic [58]
	AudioMNIST	UTS	C	Signal-space [12, 154]
	SleepEDFx	UTS	C	Signal-space [12]
	MIT-BIH	UTS	C	Signal-space [12, 154, 160]
	PAM	MTS	C	Signal-space [57, 89, 123, 175]
	HAR	MTS	C	Signal-space [110]; Pattern-space [21]; Semantic [82]
	Robot Execution Failures	MTS	C	Structural [106]
	Petroleum Distillation	MTS	F	Structural [29]
	Power	MTS	F	Structural [158]
	SMD	MTS	A	Pattern-space [144]; Structural [20]
	SMAP	MTS	A	Pattern-space [144]
Industry	SWaT	MTS	A	Structural [20]
	PSM	MTS	A	Structural [20]
	NASA Shuttle	MTS	A	Semantic [109]
	Real-world Full Flight	MTS	C	Structural [55]
HPC	ETT	MTS	F	Pattern-space [112, 113]; Semantic [94, 157]
	WSD	UTS	A	Semantic [88]
	Industrial Data	MTS	A	Pattern-space [143]
	HPAS	MTS	C	Counterfactual [9]
HPC	Taxonomist	MTS	C	Pattern-space [9]
	Cori	MTS	C	Pattern-space [9]

explanations. Since the ground-truth relevance or segmentation is known, such datasets can also support rigorous benchmarking of faithfulness, sparsity, and robustness metrics, serving as a solid basis for experimental comparison.

Real-world Datasets. For real-world scenarios, Table 7 presents datasets from domains like healthcare [68, 90], environmental monitoring [39, 156], and high-performance computing (HPC) [9]. These datasets capture a variety of temporal behaviors: they include physiological and clinical measurements, industrial system logs, and environmental observations, and are widely used for time series classification, forecasting, and anomaly detection in explainability research. In addition, Table 8 lists datasets from socioeconomic and broader application areas, such as financial time

Table 8. Summary of real-world datasets. Abbreviations: UTS = Univariate Time Series, MTS = Multivariate Time Series, Multi = Multi-modal Time Series, C = Classification, F = Forecasting, A = Anomaly Detection, I = Imputation, O = Open-Ended Reasoning.

	Data	Data Type	Task	Explanation Level
Finance	S&P 500	MTS	C	Semantic [75]
	Nikkei 225	MTS	C	Semantic [75]
	Raw Material	Multi	C	Semantic [61]
	M3	UTS	F	Pattern-space [115]
	M4	UTS	F	Pattern-space [115]
	NASDAQ	MTS	F	Signal-space [39]; Semantic [94, 157, 174]
	Volatility	MTS	F	Signal-space [85]
	Gold Price	MTS	F	Signal-space [179]
	Tesla Stock	MTS	F	Signal-space [179]
	Exchange	MTS	F	Signal-space [156]; Pattern-space [112, 113]
	NASDAQ-100	Multi	F	Semantic [174]
	KPI	UTS	A	Pattern-space [144]; Semantic [88]
Transportation	Yahoo	UTS	A	Pattern-space [144]; Semantic [31, 88]
	Linear System	UTS	C	Structural [106]
	Cruise Control	UTS	C	Structural [106]
	TOURISM	UTS	F	Pattern-space [115]
	Traffic	MTS	F	Pattern-space [158]; Structural [158]; Semantic [50]
	Nottingham	MTS	F	Semantic [58, 59]
	Singapore	MTS	F	Semantic [59]
	Glasgow	MTS	F	Semantic [59]
Multi-domain	CATraffic	Multi	F	Semantic [50]
	UCI Repository	UTS	C	Structural [106]
	UCR Archive	UTS	C	Signal-space [89]; Pattern-space [28, 150, 164]; Structural [163]
	UEA Archive	MTS	C	Pattern-space [52]
	TIMERBED	Multi	C	Semantic [177]
	Darts	UTS	F	Semantic [42]
	Monash	UTS	F	Semantic [42]
	Informer	MTS	F	Semantic [42]
	CiK	Multi	F	Semantic [7]
	TSQA	U/MTS	C/F/I/A/O/QA	Semantic [70]
	RATs40K	Multi	A	Semantic [70]
	TSB-UAD	Multi	A	Semantic [31]
	TS-Instruct QA Gold	Multi	QA	Semantic [124]

series [39, 115]. Real-world datasets introduce practical challenges such as multivariate interactions, noise, irregular sampling, and missing values [62, 138]. These make them important for studying how explanation methods perform under practical conditions. The collection of datasets also shows the variety of time series explanation tasks. In healthcare, sensor-based systems, and industrial monitoring, classification and anomaly detection are prevalent. These applications need explanations that can identify critical temporal patterns relevant to clinical conditions or signs of equipment failure. Environmental and socioeconomic datasets are more commonly used in forecasting tasks, where explanations focus on temporal trends and causal dependencies between different variables.

Multimodal Datasets. In addition, the growing availability of multi-modal datasets [31, 124] has provided external information that enriches time series tasks. In healthcare, *Test-Positive* and *Mortality* [61, 75] combine physiological measurements with clinical text, enabling models to relate temporal dynamics to higher-level medical information. Similarly, *ECG-Bench* [73] and *PTB-XL* [13] associate ECG time series with detailed textual annotations or question-answer labels. For environmental data, *New York* [61] links time series with textual or spatial descriptors of urban conditions, allowing explanations to situate temporal variations within relevant events or regional characteristics. In the financial domain, *Raw Material* [61] and *NASDAQ-100* [174] supplement temporal signals with textual metadata and descriptive attributes of the associated commodities or companies. In transportation, *CATraffic* [50] introduces a structural modality by embedding traffic measurements within a fixed road-network graph.

4.2 Applications

This section discusses how time series explanation is applied in real-world scenarios and how different domains need and use explanations at different cognitive levels. In particular, time series explanations are important to high-stake applications (e.g., healthcare), interdisciplinary communications (e.g., environmental science), and industrial scenarios (e.g., manufacturing systems).

In healthcare, applications such as patient monitoring, clinical risk prediction, and ICU event forecasting need explanations to support clinical validation. At the signal-space level, models [22, 68, 140] highlight the influential visits, variables, and time steps related to the patient. Building on this, pattern-space methods [77, 112] identify recurrent segments or representative trajectories. At the structural level, rule-based and counterfactual frameworks [116, 168] capture dependencies among physiological variables for causal reasoning. At the top of the hierarchy, TimeCAP [75], TimeXL [61], and GEM [73] integrate LLMs to generate textual explanations that describe disease progression and treatment response. Time series explanations have progressed across all four cognitive levels, reflecting the diverse needs of clinical decision support.

In environmental science, explainability is essential for validating climate models, understanding pollution patterns, and guiding policy choices. At the signal-space and pattern-space levels, methods [57, 85, 156] identify influential time steps and common trends in air quality and energy forecasts. Structural-level approaches [106, 125, 158] show relationships among meteorological and environmental factors, providing causal insights. At the semantic level, methods [58, 94] combine multi-modal data and LLMs to turn forecast results into natural language explanations.

Industrial and manufacturing applications rely on explainable time series models for fault diagnosis, predictive maintenance, and process optimization. A trend is the focus on higher-level reasoning. Most studies concentrate on pattern, structural, and semantic explanations instead of low-level signal attribution. At the pattern-space level, methods [112, 113] extract representative temporal segments that capture operational cycles or abnormal dynamics. Structural-level approaches [55, 143] use causal graphs to trace how faults spread and evaluate the effects of interventions. Finally, semantic-level frameworks [88, 94, 157] use LLMs or RAG to create human-readable explanations that outline system states and performance anomalies. This helps connect complex model outputs with useful insights for operators.

In transportation systems, effective traffic flow forecasting and event prediction rely on the ability to explain complex traffic dynamics, inform demand forecasting, and support mobility planning. At the pattern-space level, N-BEATS and Basisformer [112, 115] identify recurring patterns in traffic flow and travel demand. Structural-level approaches [106, 158] use rule-based systems or causal graphs to reveal dependencies among different areas and transportation modes. Finally, at the semantic level, LLM-based frameworks [50, 58, 59] create natural language narratives that explain the causes of congestion, evolving trends, and regional interactions.

5 Challenges and Future Directions

Building on the cognitive perspective introduced in Section 3 and the domain applications discussed in Section 4, this section highlights several open challenges in time series explanation and suggests possible directions for future work.

5.1 Improving Explanation Methods

Most existing methods focus on a single cognitive level. Signal-space and pattern-space explanations identify important timesteps or segments with high **precision**, but they are limited in **readability**. Structural and semantic explanations are

easier for humans to follow, but they lack quantitative metrics for evaluation. Bridging these levels remains a significant challenge in practice.

Cross-Level Integration. A valuable direction is to design frameworks that can generate explanations from multiple cognitive layers while keeping the outputs consistent. Explanations that focus on certain time points should support the patterns identified at the pattern-space level, which in turn should match the causal structures and semantic explanations created at higher levels. Achieving this consistency across levels would make explanations more accurate and easier for humans to understand.

Multi-Modal Time Series Explanation. Many time series encountered in real-world settings come with additional forms of information, such as text, event sequences, images, and structured knowledge [60, 135]. Methods that can process these sources together may provide more accurate explanations. For example, physiological signals can be paired with clinical notes in healthcare [95], and sensor readings can be linked to maintenance logs in industrial monitoring [139]. However, existing multi-modal time series explanation methods rely on simple fusion strategies, in which different modalities are combined before explanation. Such designs often reduce temporal information to coarse representations, making fine-grained temporal evidence unavailable during reasoning. As a result, it is difficult to identify how multiple modalities jointly contribute to an explanation.

Explanations for New Architectures. Recent advances in time series foundation models [84], state space methods [43], and large pre-trained temporal encoders bring new challenges for explainability. These models have complex internal mechanisms that existing methods cannot address. Developing explanation techniques suited to these new architectures, such as probing latent states in Mamba-style models [43] or explaining cross-modal attention in time series foundation models, or use knowledge distillation to explain large models with smaller explainable models [15] is a critical area.

Scalability and Efficiency. Many explanation methods require significant computational resources [11, 24], which limits their use in long sequences, high-dimensional settings, or real-time applications such as continuous monitoring of complex systems or physiological signal analysis. Research on approximation methods and online explanation procedures would broaden the range of situations where explainability tools can be practically used.

5.2 Evaluation and Benchmarking

The lack of standardized evaluation criteria is a barrier to developing explainable AI in time series analysis. There are quantitative metrics for signal-space and pattern-space explanations, including deletion tests [32, 57], insertion scores [68], and localization accuracy [30, 37]. However, these metrics fail to apply to structural and semantic explanations. Textual explanations often rely on human judgments [70, 88] or LLM-based scoring [72], which introduces subjectivity and variability.

Developing Standardized Benchmarks. The community would benefit from standardized benchmark datasets that provide ground-truth explanations for all four cognitive levels. Benchmarks should cover various domains, such as healthcare, finance, industrial, environmental. And they should also include different task types. A standardized benchmark suite would enable more consistent evaluation and comparison of methods.

Grounding Semantic Explanations. A valuable evaluation strategy is to map semantic explanations back to lower-level evidence. If a generated textual explanation can be traced to specific time steps, subsequences, or structural relationships, we can assess its validity with existing quantitative metrics. This grounding approach would lessen the need for subjective evaluation while preserving the depth of semantic explanations.

Faithfulness-Plausibility Trade-offs. Evaluations need to balance between faithfulness (how accurately an explanation reflects the model’s decision process) and plausibility (how convincing an explanation appears to humans). Methods that optimize for one aspect may sacrifice the other. Understanding this trade-off is important for the practical use of explanations.

Robustness and Reliability. Semantic explanations produced by LLMs can be affected by how prompts are phrased, the order of variables, and formatting choices [4, 44]. Future evaluation protocols should measure this variability and develop methods to identify hallucinated or unsupported statements in generated explanations.

5.3 Explanation-Assisted Time Series Learning

Current research on explainability mainly treats explanations as outputs that come after model training [101, 150]. However, explanations contain valuable information about which temporal patterns are important for the task. Integrating explanations during training may improve model robustness, representation quality, and decision-making.

Explanation-Guided Contrastive Learning. In graph learning, explanation methods have been successfully used within contrastive frameworks [16], where explainers find task-relevant subgraphs to help guide augmentation strategies. Similarly, time series explanation methods can identify notable patterns that should be preserved during augmentation. Current contrastive frameworks [34, 38, 92] create augmentations through random crops, masking, or temporal distortions without considering task relevance. Explanation-aware augmentation could produce more meaningful and distinctive representations.

Self-Explaining Models. Instead of using post-hoc explanations, future work could optimize both prediction accuracy and explanation quality simultaneously. These self-explaining models may take explanations as an intrinsic part of the prediction process, creating a better connection between the model’s reasoning and its explanatory output. This shift from “explain after” to “explain during” could lead to more accurate and faithful explanations.

5.4 Human-AI Interaction and Trust

The main goal of explainability is to help humans understand, make decisions, and trust AI systems [97]. However, limited research has examined how humans actually interpret, use, and trust time series explanations.

User Studies for Different Expertise Levels. Preferences for explanation types vary based on user background. Domain experts need structural or pattern-level explanations that relate to their knowledge, while general users might prefer simpler, high-level explanations. We need systematic user studies to characterize these preferences and to help design flexible explanation interfaces.

Interactive Explanation Systems. Static explanations provide limited support for complex time series analysis. Interactive systems that allow users to ask questions about an explanation, explore perspectives, or examine the model’s reasoning can improve the usefulness of explanations [103].

Personalized and Adaptive Explanations. Future explanation systems should adjust their outputs based on user characteristics, such as domain background and task requirements. A clinician reviewing patient monitoring data has different explanation needs with a data scientist debugging a forecasting model. Personalization systems could adjust the explanation level to make explanations easier to use in various situations.

Trust Calibration. Collaboration between humans and AI relies on trust. Users need to believe explanations when they are reliable, but also recognize their limitations when they are not. More research is needed to understand how the

explanation quality affects trust and how uncertainty should be communicated. Overconfident explanations may cause bias, while overly cautious presentations might prevent adoption [153].

6 Conclusion

This survey provides a systematic overview of time series explanation from a human-cognitive perspective. It structures existing methods into four levels: signal space, pattern space, structural, and semantic. It describes how explanations develop from low-level signal attribution to high-level semantic reasoning. The survey starts by reviewing signal-space methods that find important time steps and features. Next, it discusses pattern-space methods that identify meaningful temporal segments, prototypes, and counterfactual trajectories. It then examines structural explanations that reveal model decision-making through rules, causal graphs, and event relations. Finally, it emphasizes the emerging semantic level, where large language models produce natural language explanations that convey temporal reasoning and contextual understanding. Additionally, the survey points out several open challenges, such as integration of multi-level reasoning, the lack of unified evaluation metrics, and the need for frameworks that are scalable and focused on humans. Future research may focus on building cognitively consistent and multi-modal explainability systems that connect algorithmic transparency with human understanding.

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